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Navigating the Energy Transition: The Impact of Delayed Hydroelectric Projects on the Democratic Republic of the Congo's Energy System based on MESSAGE-SPLAT tools

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Abstract. The Democratic Republic of the Congo (DRC) faces energy supply challenges exacerbated by delays in hydroelectric projects. This case study explores the potential consequences of delaying the INGA III hydroelectric project by 2035, a critical component of the nation's energy strategy. Through comprehensive modelling using MESSAGE - SPLAT tools, we analyze the ripple effects on the DRC's energy system. Our findings underscore the importance of timely execution of hydro projects to avoid increased reliance on non-renewable sources and heightened emissions. Delaying INGA III leads to an energy deficit, necessitating the use of natural gas and diesel power plants. Additional capacity of 510 MW needed between 2025 and 2035. Greater reliance on fossil fuels, with natural gas peaking at 16% of the energy mix by 2030. Battery storage emerges as a crucial component to balance demand and supply. Timely execution of hydro projects is vital for the DRC to maintain a sustainable and cost-effective energy strategy. Policymakers must consider the broader implications of project delays and prioritize investments in complementary renewable technologies, like hydrogen storage feasibility.

Keywords: DRC's energy system, MESSAGE-SPLAT, Hydropower, Grand Inga, Battery storage

1. Introduction

The DRC's energy landscape faces significant challenges, particularly in managing energy supply amidst project delays. The DRC possesses enormous renewable energy potential, primarily through the Grand Inga hydropower complex, which, if fully realized, could become the largest hydroelectric power generating facility in the world. However, the development of the Grand Inga project faces significant political, financial, and geopolitical challenges that have delayed its implementation, impeding the country's ability to resolve its severe energy crisis. Notably, the suspension of international funding and the complexity of coordinating regional power needs have created a renewable energy conundrum,



limiting progress on this transformative infrastructure despite its promise to support economic growth and regional electrification. Addressing these challenges with increased transparency, political stability, and innovative financing mechanisms is crucial for unlocking the sustainable energy future envisioned by the Grand Inga initiative [1].

Since 2021, the DRC's National Energy Policy (PNE) has served as the cornerstone of its energy strategy, fully aligned with the Government Action Program (2021–2023) and the broader National Strategic Development Plan (PNSD) [2]. Anchored in a vision to make energy a driver of sustainable development by 2040, the PNE seeks to attract local and international investment by streamlining public-sector procedures, strengthening legislative and regulatory frameworks, and improving the overall business climate. Its objectives include closing the energy access gap through the sustainable use of domestic resources, securing supply and national sovereignty, and promoting efficient, affordable technologies. The PNE also emphasizes research and development, sound governance, decentralization and environmental stewardship throughout energy production and consumption. In line with its Paris Agreement commitments, the DRC has submitted Nationally Determined Contributions aiming to cut greenhouse-gas emissions by 21 percent by 2030 — 2 percent unconditionally and a further 19 percent contingent on international support — thereby embedding climate action into its development agenda. Altogether, these measures are designed to modernize the power sector, strengthen its resilience and ensure it contributes to the country's wider sustainable development objectives [3]. Despite these efforts, biomass - mainly charcoal and firewood - remains the dominant energy source in the DRC, accounting for 94% of total consumption [4], while electricity remains marginal and is concentrated mainly in the industrial sector in the south, where mining is most developed.

Since 2021, the Congolese government has allocated a major portion of its national budget to the deployment of isolated mini-grids in rural areas, in order to reduce the electricity distribution gap between urban and rural zones. This effort mobilizes a total of USD 41.8 million (approximately USD 16.8 million per year) and represents more than 80% of public energy funding [5]. While it contributes to the country's electrification, it provides only limited support for strengthening the national interconnected grid.

As part of the Mission 300 Initiative, launched in April 2024 by the World Bank Group and the African Development Bank, the DRC has pledged to increase its electricity production capacity to 11,000 MW and achieve a 62% electricity access rate by 2030, giving priority to renewable energies, particularly solar and hydroelectric. To this end, studies on the Inga site have proposed sequential deployment of six new power plants (Inga 3 to Inga 8), with the first phase - Inga 3 - providing 4,800 MW within the Grand Inga project, whose final capacity is expected to peak at 44,000 MW to strengthen the national electricity grid and regional interconnexion [6].

Recent advances in energy system modelling, particularly using tools like OSeMOSYS, have enabled scenario-based planning that integrates technical, economic, and environmental factors to optimize resource use. A hydro-focused scenario projects that by 2050, DRC could achieve an installed capacity of around 23 GW dominated by hydropower, significantly enhancing electrification and reducing carbon emissions by over a gigaton. However, realizing this vision requires substantial investments—estimated at around USD 100 billion—and strategic policy interventions that prioritize decentralized electrification and grid expansion. This research contributes a pragmatic roadmap for the DRC's energy transition, emphasizing the pivotal role of hydropower while underscoring the need for complementary renewable technologies and robust energy governance aligned with climate objectives [7].

Therefore, delays in implementation of these hydro projects have significant implications for the country's energy security and environmental goals. This paper presents an analysis of these potential impacts, offering insights into alternative strategies and their feasibility.

2. Methodology

2.1. Application of MESSAGE – SPLAT tools for Energy supply modelling

The MESSAGE-SPLAT tools are a modelling framework developed and applied within the African continent for energy supply analysis and capacity expansion optimisation [8]. SPLAT (System Planning Test) is an Excel-based utility interface designed to interact with the MESSAGE energy system model optimising energy supply systems (figure 1). The framework supports least-cost capacity expansion optimisation considering diverse energy resources and technologies across African countries and regions.

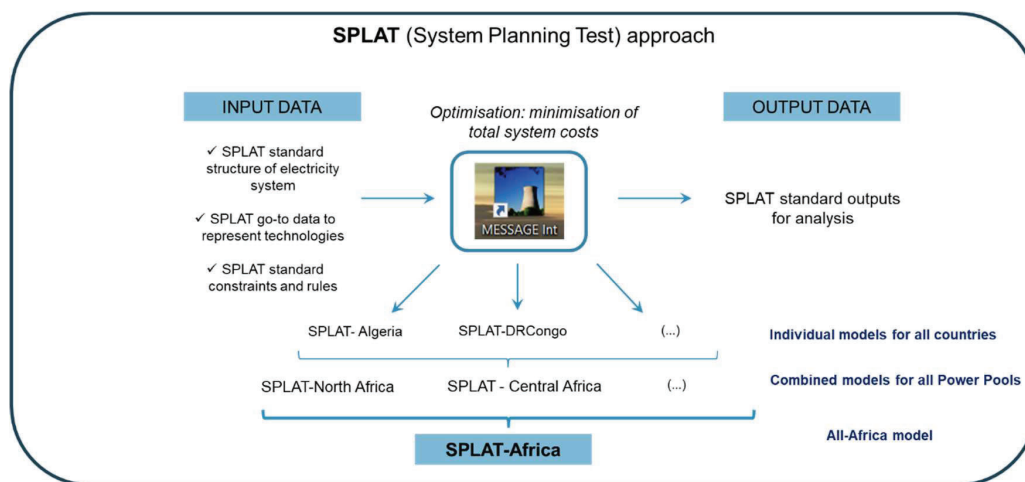


Figure 1. SPLAT approach [8] [Authors]

SPLAT operates as the interface managing data flows to/from MESSAGE, allowing modifications to technologies, constraints, and time resolution within an Excel environment. The optimisation objective function is the minimisation of total system costs including capital investment and operational expenses (investments costs annualized and all costs discounted to base year), under several technical and temporal constraints. Constraints include firm capacity, renewable penetration limits, build rates, and minimum utilisation rates to ensure system reliability. The Excel-based SPLAT interface provides interaction capabilities with MESSAGE, bulk data updates, scenario management, and outputs visualisation, enabling detailed planning for interconnected power systems [8].

2.2. Data and Inputs

The SPLAT-Africa model incorporates country-level databases of resources, existing technologies, power plants, storage schemes, and transmission infrastructure. Costs, fuel inputs, lifetime, and temporal-spatial variability are integral data elements for each technology option.

The data inputs used in the Democratic Republic of the Congo (DRC) MESSAGE-SPLAT modelling presented include several key components related to the country's power system and future scenario planning. The inputs comprise detailed descriptions of the existing and candidate power plants, transmission lines, and electrification using country default parameters from the African Continental power systems Masterplan's SPLAT – CMP model 2023, as follows:

- The current installed capacity of 3.2 GW, and the available power capacity of 2.17 GW within the DRC system, provides a foundation for modelling supply capabilities [9].
- Existing power plants include 18 hydroelectric plants, 6 diesel plants, and 1 natural gas plant. Candidate power plants planned for development include 5 hydropower plants (including INGA

III and others), with a total potential capacity of 19.4 GW. Committed projects such as a 160 MW hydro plant are also factored in.

- Demand and supply profiles of energy forms are considered for variability across time of day (hours), time of 3 seasons. This is modelled and simulated in MESSAGE by subdividing a year into 3-time blocks, as load regions.
- The development of isolated mini-grids was not taken into account in the model.

The main constraints are as follows: resource availability is set at 100% for fossil fuels, pumped storage, hydropower and geothermal, 0% for variable renewable energy (VRE), and 75% for battery storage accounting for degradation and grid stability reserves. Instantaneous VRE penetration is capped at 70%, while the annual build-out limit is restricted to a maximum of 25% of the previous year's peak demand. In addition, minimum annual capacity utilization is enforced: 5% for diesel, heavy fuel oil and open-cycle gas, 20% for coal, and 15% for combined-cycle gas.

Two main scenarios are modelled: the Reference scenario assumes INGA III is commissioned by 2030, while the DelayedHydro scenario assumes a delay until 2035. These scenarios are designed to assess the impact of project timing on energy capacity, emissions, and system costs. The Reference scenario from SPLAT – CMP model 2023 where INGA III starts by 2030, and the DelayedHydro scenario where INGA III is delayed until 2035, for individual model without considering interconnected power systems. We used projected technology costs from the SPLAT-CMP database and a conservative fuels price trajectory. The scenarios specify analysis periods (up to 2040) and assumptions about project timelines and capacity additions.

3. Results and discussion

The analysis of the Democratic Republic of Congo's power system under two scenarios—the Reference scenario with the INGA III hydropower plant online by 2030 and the DelayedHydro scenario with INGA III delayed until 2035—reveals important insights.

In the Reference scenario, the installed capacity (figure 2) is expected to increase through the addition of INGA III and other hydro projects, supplemented by battery storage providing about 102 MW of additional capacity (figure 3). Conversely, in the DelayedHydro scenario, a delay in INGA III commissioning would require alternative capacity, mainly through the deployment of approximately 510 MW of natural gas power plants along with additional storage batteries to fill the gap.

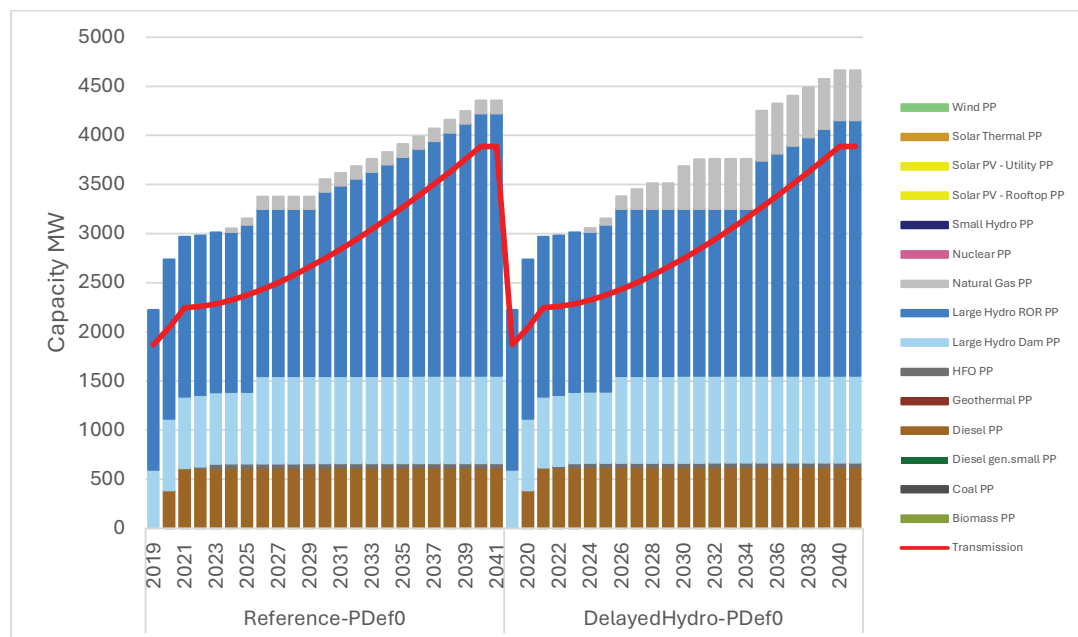


Figure 2. Total annual installed capacity [Authors]

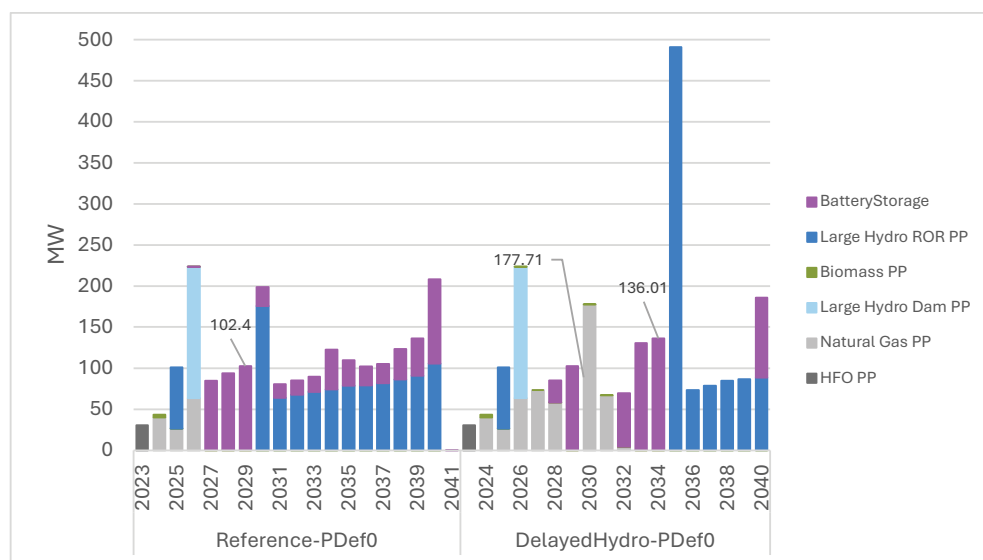


Figure 3. Additional annual capacity [Authors]

Regarding the energy mix (figure 4), the delay in the hydro project results in a substantial increase in fossil fuel usage. While fossil fuels constitute less than 10% of the generation mix in the Reference scenario, their share could rise to nearly 20% up to 2035 in the DelayedHydro scenario. This shift leads to a significant increase in carbon dioxide (CO₂) emissions, with the delay causing emissions to rise by approximately 42%, or an additional 4,672 million tons of CO₂ relative to the Reference case (figure 5). Yet, The DRC's annual GHG emissions are estimated to 677 MtCo₂e, representing 0.49% of the global world emissions [10] [11].

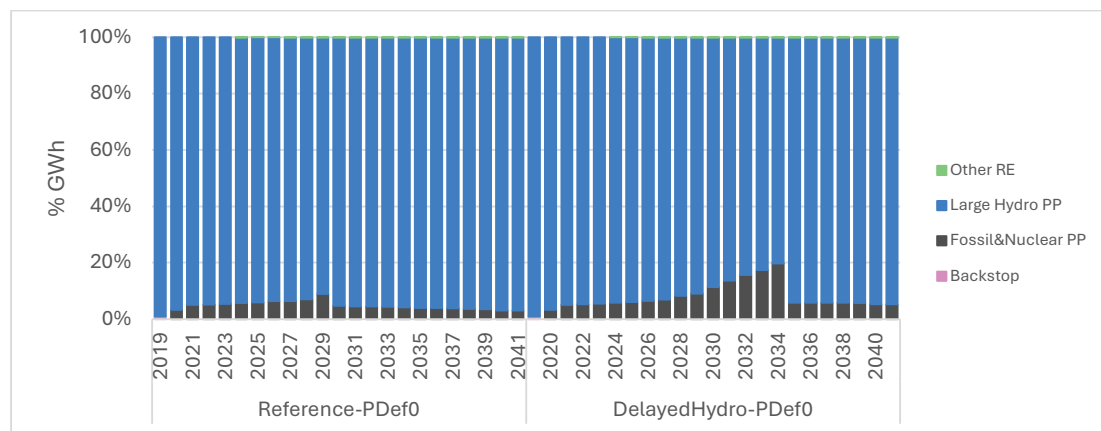


Figure 4. Annual electricity production shares [Authors]

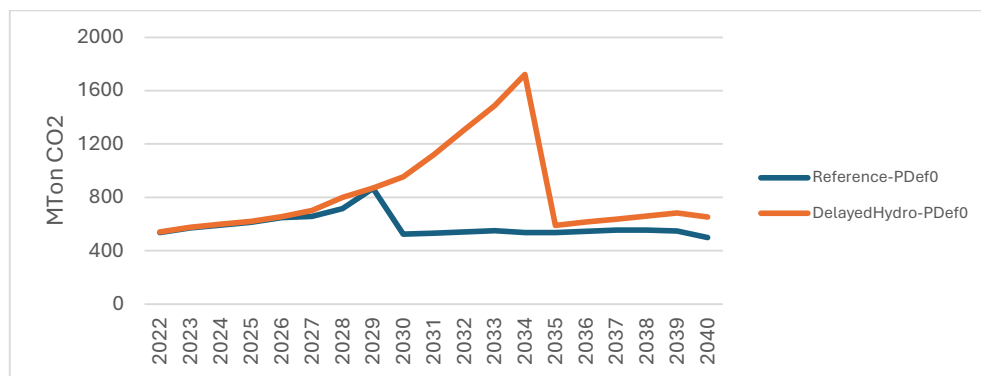


Figure 5. CO₂ Emissions [Authors]

Economically, the delay in INGA III also impacts investment and operational costs. The cumulative investment costs necessary to meet electricity demand are estimated to be higher by about 187 million USD if the hydro project is postponed to 2035 (figure 6).

Moreover, the average annual system costs in the delayed scenario increase by around 3%, compared with timely development of the hydropower assets.

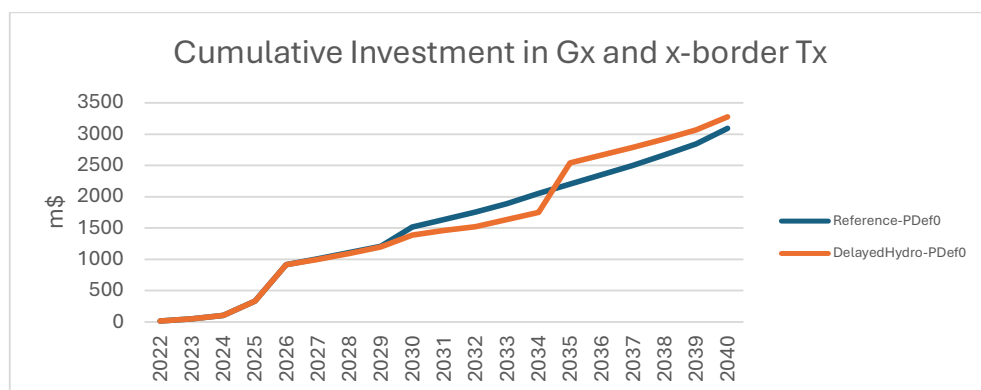


Figure 6. Cumulative investments [Authors]

From the system operation perspective, hydropower remains the primary baseload generation source in both scenarios. However, diesel and natural gas power plants are increasingly relied upon to meet peak demand, especially during night hours and seasonal peaks. Examining the hourly electricity production for 2029 (figure 7) and 2040 (figure 8), hydropower consistently serves as the baseload generation in both scenarios. However, diesel and natural gas power plants are primarily used to meet peak demand during the last five hours of each day, especially during the third season of the year. In 2029, the DelayedHydro scenario sees diesel power contributing about 21% during peak times, identical to the Reference, but the latter requires higher reliance on battery storage (8%) compared to 4% in the delay scenario. By 2040, diesel and natural gas usage increases in the delayed scenario with diesel at 11% (lower than 14% in Reference) but natural gas picks up to 12%, reflecting greater fossil fuel dependency. Simultaneously, battery generation needs remain higher in the Reference scenario (14%) relative to the delayed case (11%).

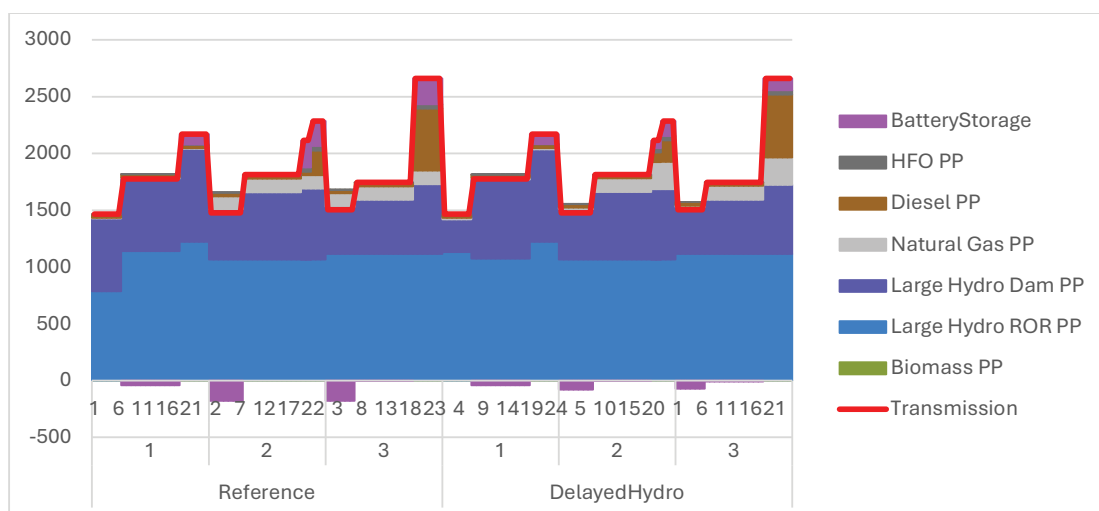


Figure 7. Annual Electricity Production profile (Output – MWh) 2029 [Authors]

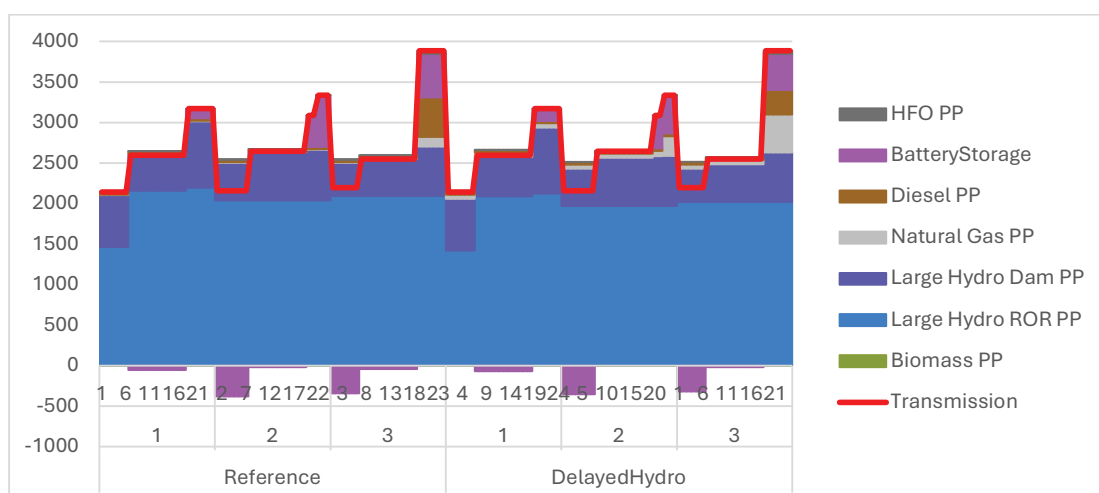


Figure 8. Annual Electricity Production profile (Output – MWh) 2040 [Authors]

The analysis highlights the critical role of battery storage in both the Reference and DelayedHydro scenarios as a key complementary technology to hydropower in meeting DRC's electricity demand. In the Reference scenario, storage batteries provide an important share of additional capacity—guaranteeing around 102 MW—that helps optimize the integration of renewable hydropower and supports peak demand management. In contrast, when the INGA III project is delayed until 2035, the reliance on storage increases even further, with batteries required to guarantee 136 MW alongside greater deployment of natural gas power plants. This underscores storage energy as a vital component for grid flexibility and stability.

However, limitations such as static fuel prices, hydrological variability, and lack of regional interconnection modelling may influence the results and should be considered in future studies, especially during peak hours or seasonal fluctuations when hydropower alone cannot fully meet demand. Although postponing INGA III increases the fossil-fuel share of electricity generation and raises both investment and operating costs, battery storage can offset some of these drawbacks by cutting reliance on expensive, high-emission diesel and gas plants during peak demand.

While INGA III contributes significantly to the generation capacity and system cost reduction, the results emphasize the necessity for additional complementary hydropower projects. This is especially true given that not all energy from INGA III is intended solely for local consumption [12]. Developing other hydro plants will be critical to minimizing reliance on fossil fuel generation, even under the Reference scenario.

4. Conclusion

Delaying the INGA III hydropower project requires rethinking system strategies and introducing complementary technologies to meet electricity demand. Unfortunately, these complementary sources are mostly non-renewable, increasing both economic costs and environmental impacts.

The modeling results reveal that postponing the INGA III project until 2035 would cause an energy deficit within the system. To meet growing demand in this delayed scenario, it would be necessary to deploy additional capacity primarily from natural gas power plants, with around 510 MW required between 2027 and 2040. In the Reference scenario, the additional capacity needed can be largely supplied by battery storage with an estimated guarantee of 102 MW, whereas in the delayed case, an extra 178 MW of natural gas power plants and 136 MW of storage batteries become necessary by 2030. The analysis emphasizes the need for the DRC to plan and develop additional hydropower plants alongside INGA III to minimize dependence on diesel and natural gas power plants. This strategy remains critical even if the INGA III project proceeds as scheduled because energy generated from INGA III will not be solely consumed within DRC but also exported regionally. Additionally, the results emphasize that timely development of INGA III combined with substantial battery storage deployment will optimize system costs, reduce carbon emissions, and enhance electricity reliability. However, in the event of delays, expanding storage capacity alongside flexible fossil fuel plants becomes indispensable, though less desirable due to cost and environmental effects.

Therefore, moving forward, the DRC should prioritize not only the on-schedule completion of INGA III but also investments in other hydro plants and energy storage technologies as part of a holistic strategy to modernize the power system. While battery systems provide significant support for peak demand management and renewable integration, their scalability, duration, and resource limitations call for complementary solutions. One promising avenue is the development of green hydrogen technologies, which offer long-duration energy storage with flexibility for seasonal variations and large-scale balancing.

Future work should focus on assessing the techno-economic feasibility of green hydrogen production, storage, and utilization within the DRC context. Broader research is needed to understand the long-term applicability of the methodology across other developing countries, storage, and

utilization within the DRC context. Demonstration projects, policy frameworks, and infrastructure development will be crucial to realize the potential of hydrogen as a complementary storage solution. Additionally, combining hydrogen with advanced battery technologies can provide a diversified energy storage portfolio, enhancing system flexibility and reliability as DRC expands its renewable capacity.

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“The authors used ChatGPT to check grammar issues and to improve readability. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.”

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